SignalDNA

Neural networks from first principles | Synthetic transient detection

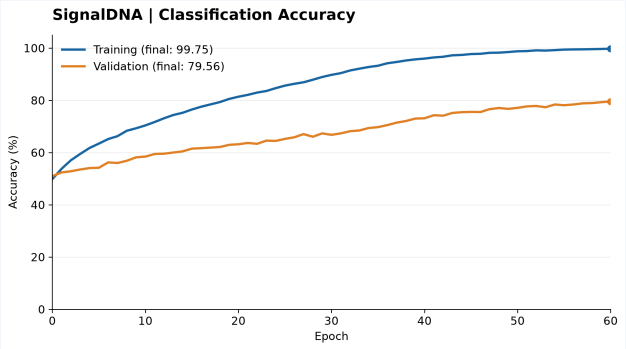
78.78%
Held-out accuracy

0.8613
Held-out ROC-AUC

67,713
Parameters

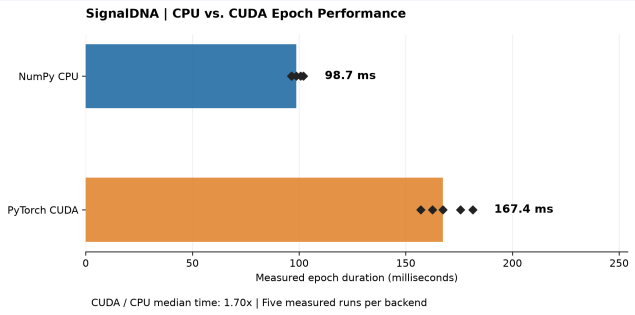
Original NumPy network: 1024 inputs, 64 and 32 ReLU units, 1 sigmoid output. Trained on balanced synthetic transients; 12,000 signals, 1,800 held out. Motivation: learning AI before exploring embedded and FPGA applications.

TRAINING & GENERALIZATION



60 epochs; train vs. validation

CPU VS CUDA



Five warmed-up runs; float64, batch 64

Train 99.75% | Validation 79.56% | Held-out 78.78%

Median epoch: CPU 98.72 ms | CUDA 167.37 ms | GPU/CPU 1.70x

WHAT THE EXPERIMENTS SHOW

Training accuracy greatly exceeded validation accuracy: generalization remains the central challenge. CPU and CUDA metrics matched at 61 archived checkpoints (maximum absolute difference 2.22e-16); the CPU was faster for this specific small workload. CUDA timings include synchronization and CPU transfers.

LIMITATIONS & NEXT STEPS

Synthetic signals only; one principal architecture and seed; one locked held-out evaluation. No validated classical DSP baseline, real-sensor testing, or FPGA deployment. Next: classical baselines, regularization, multiple seeds, real data, and hardware inference measurements.